TRAIN SIM CLASSIC · VEHICLE WORKFLOW GUIDE

# Part 6

## Build, test and ship

A short iteration loop, a test drive that catches problems in the right order, ways to inspect what the game really loaded, and a symptom-to-fix table for the whole pipeline.

### In this part


1. The iteration loop: what to redo after each kind of change
2. Getting the vehicle into the game
3. The test drive, in order
4. Looking inside compiled files
5. Shipping
6. Troubleshooting: symptom → cause → fix


Steps inside the game's editors vary slightly between TSC releases; where exact menu names matter they are marked **check**.

# 1 · The iteration loop

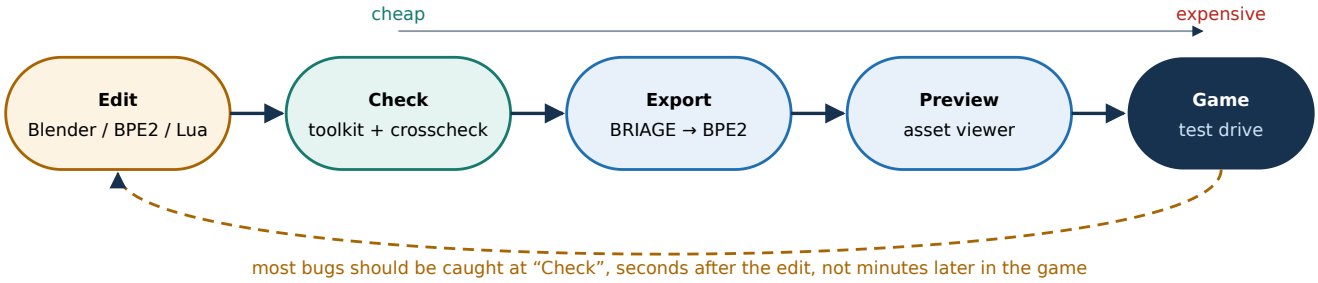



**Figure 6.1 · Cost grows to the right.** Move every check as far left as it will go.

| You changed… | Redo |
|---|---|
| Mesh shape, UVs, materials, textures | BRIAGE IGS export → BPE2 export of the vehicle |
| Object names or hierarchy | IGS **and every .ia** (names must match) → re-check blueprint links (crosscheck) → BPE2 export |
| An animation | That .ia only → BPE2 export; clear the preview cache if it looks stale |
| Bogie/wheel/coupler position | IGS → toolkit Measure + Patch → BPE2 export |
| Blueprint values only | BPE2 export |
| Lua only | BPE2 export (compiles `.out`); reload the scenario |
| CSV curves | BPE2 export of the simulation blueprint |

> **Close the game before exporting**
>
> Exporting into a provider/product the route editor has loaded can corrupt the route on save (BRIAGE §9.1.3). Many changes also only take effect after the vehicle is re-placed in the scenario (BRIAGE §10.6).

# 2 · Getting the vehicle into the game

1. Export from BPE2 so the compiled files exist under `Assets\Provider\Product`.
2. Start TSC, open a route in the editor, and in the object set filter enable your **Provider** and **Product** check (menu names vary by release).
3. Drag the engine (or wagon) from the rolling-stock browser onto the track. Build a short train with a matching-brake wagon.
4. Save as a test scenario (Free Roam or Standard) and drive it. Keep one test scenario per vehicle; re-place the vehicle after blueprint changes.
5. Optional: a **consist blueprint** (loco + wagons) makes the train reusable in scenarios check Quick Drive listing rules on your build.

# 3 · The test drive, in order

Test in this order: each step depends on the previous one, so a failure early saves you chasing symptoms later.

| # | Check | If it fails, look at |
|---|---|---|
| 1 | Vehicle appears, textured, right way round | Geometry path, texture paths, front = +Y |
| 2 | Sits on the rails, not floating/sunk | Origin on rail head, wheel radius = centre height |
| 3 | Couples to the wagon without a gap | Coupling pivot/receiving matrices, coupling blueprints |
| 4 | Spawn state: brakes released, controls at sensible values | DefaultValue of each control |
| 5 | Brakes charge and release | Air system / ejectors, compressor, pressures |
| 6 | Moves on throttle; wheels spin at the right speed | Curves vs MaxForce, Powered, axle NodeIds, radius |
| 7 | Bogies turn on curves, not ahead of or behind the mesh | BogeyPivotX vs mesh |
| 8 | Every cab control clicks and moves with its value | PickTransformName, .ia range, linear keys |
| 9 | Exterior animations (pantograph, wipers, doors) | AnimSet IDs, AnimStyle, node merging |
| 10 | Gauges read correctly across their range | Control min/max vs dial calibration |
| 11 | Lights by direction and at night; numbers display | Light children, light meshes, digits set |
| 12 | LODs switch cleanly at distance | LOD distances, hidden LOD twins |
| 13 | Script features; LogMate shows no errors | Part 5 |
| 14 | Reverse direction, other cab end, AI driving | Double cab, MaxPermissibleSpeed, GetIsPlayer gating |

# 4 · Looking inside compiled files

TSC ships a converter, `serz.exe`, that turns compiled `.bin` blueprints into readable XML and back. The raw geometry dumps supplied with this project (`cHcGeometry` XML with `TransformName`, `SourceLToPTransform`, LOD controllers and materials) were produced by a conversion of this kind. Use it to:

- confirm what BPE2 actually compiled (values, paths),

- list the node names in an exported model and feed them to `tsc_crosscheck.py` (`--geo`, `--cabgeo`),
- study how stock vehicles solve a problem you are facing.

```
serz.exe "Assets\MyProvider\MyProduct\...\MyLoco.bin"   → MyLoco.xml
python tools\tsc_crosscheck.py --engine MyLoco.xml --geo MyLoco.geo.xml --cabgeo
MyLoco_cab.geo.xml
```

**check** serz's exact command-line options on your install; it lives in the RailWorks folder.

# 5 · Shipping

- Ship the **Assets** files your vehicle needs (blueprints `.bin`, geometry, textures, animations, curves, `.out` scripts, sounds). Include Source only if you want others to modify it.
- Package with the TSC package manager (`.rwp`) or a zip that mirrors `RailWorks\Assets\Provider\Product`. **check**
- List dependencies (other products your blueprints reference, e.g. Kuju coupling or particle blueprints).
- Do not redistribute Kuju/DTG sample files as your own; re-author anything copied from samples.
- Ship with DEBUG alerts off and `print` spam removed.

# 6 · Troubleshooting: symptom → cause → fix

## 6.1 In Blender / BRIAGE

| Symptom | Cause → fix |
|---|---|
| “No mesh or empty object found” | Nothing selected/visible for the chosen export mode → select, or unhide (BRIAGE §9.2) |
| “Please export to the same drive…” | Textures on another drive → move textures or use path tools (§9.3) |
| Wrong number of texture slots | Shader expects N slots → see the shader table in Part 2 |
| “Inconsistent LODing”, offset objects | LOD child not under LOD n−1, different base name, smaller distance, or 24-char truncation clash (§9.10) |
| Name changed in log | Missing/bad `L_DDDD_` prefix → rename (§10.4) |
| Settings seem ignored | Unapplied scale → Ctrl+A Scale, re-export with Verbose (§9.8) |

## 6.2 In BPE2 / preview

| Symptom | Cause → fix |
|---|---|
| No textures in preview | Texture path mapping mode or directory → check the IGS log (§9.4) |
| Missing frames of an animated texture | Only frame 1 is referenced → export the other frames manually (§6.4) |
| Animation doesn't play | “Animate in editor” off, merged node, name mismatch, stale cache (§9.5) |

| Symptom | Cause → fix |
|---|---|
| Export error on a Lua file | Syntax error → fix and re-export |
| File with extra dots misbehaves | BPE2 treats text after the first dot as extension → rename (§9.7) |

## 6.3 In game

| Symptom | Cause → fix |
|---|---|
| Spawns with brakes on | TrainBrakeControl default > 0 → 0 |
| Weak or no traction | MaxForce below curve peak; unpowered bogies; throttle curve not starting at 0,0 |
| Brakes never charge | No compressor / cut-out 0 / ejectors missing |
| Brake handle dead | CabControlName ≠ ControlName |
| Wheels sink or float | Radius ≠ modelled centre height |
| Bogie swings in the wrong place | BogeyPivotX ≠ mesh |
| Cab control not clickable | PickTransformName node missing in the cab IGS |
| Lamp/indicator never shows | Visibility element TransformName missing |
| Lever jumps or lags | Non-linear keys; keys not at first/last frame |
| Crash on load (newer builds) | Suspect a self-closing `<LocoBrakeAssembly />` (handoff doc) and empty required lists |
| Script feature stops mid-session | Runtime error → LogMate |
| AI drives too fast / never stops | MaxPermissibleSpeed; brake performance |
| Crash or corruption after editing | Exported while the route editor had the product loaded (§9.1.3) |